Production of Tuberculosis Vaccine in Uganda

TEAM 40

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Aim and Background

Mycobacterium tuberculosis is a disease which mainly affects the lungs, and in more severe cases, bones, abdomen or the nervous system. The disease causes fever, coughs and chest pain. Currently, a vaccine - commonly known as BCG - is used to protect people from TB. However, with the increase of antibiotic intake, the human body has developed a certain resistance, thus reducing the efficiency of BCG. Uganda, a country in East Africa, is one of the most vulnerable countries in the fight against TB, with around 250 TB cases per 100,000 population[1]. This challenge's goal is to create a new, sub-unit/protein-based tuberculosis vaccine, to offer a potential solution to the antibiotic resistance problem. The aim of this project is to design a small scale model of a bioreactor, one of the main components used in the vaccine creation process, using a plastic cup. The design will then be scaled up and applied to larger costs and quantities.

Non-technical considerations

It is important to note that, in any tech-related project, external factors should also be assessed and taken into consideration. A PESTEL analysis was conducted to assess the risk and security factors of establishing a vaccine plant in Mbarara. In terms of politics, tensions with Tanzania, Rwanda, and the Democratic Republic of Congo (DRC) were taken into account. Building the vaccine plant could push Uganda to offer some vaccine doses to Congo for example, therefore helping the country to reduce their debt to DRC. Any limitations on recent technology access will also try to be avoided, by making sure the plant is provided with the most important equipment. Several cultural and social aspects were also revised: campaigns will be made to raise awareness for people who might not want to get vaccinated due to cultural beliefs. Moreover, staff will receive additional training in case they are lacking some scientific competences. This could be done via a volunteering program, where professors from abroad can offer their expertise. If not, then workers from other countries could also be hired (despite having to pay them more). New roads might have to be built to facilitate transport of goods, which might cause problems for the locals. Furthermore, legal issues may arise due to corruption: in 2018, over 38% of the population had to bribe to access public services. This is yet again another problem which was taken into account. Finally, economical factors were taken into account too: knowing that many Ugandans live under the poverty line, the aim is to make the vaccine free of charge for the people. Surrounding environments were also taken into consideration regarding the disposal of the vaccine and ensuring it does not pollute the land, by avoiding just dumping the waste on landfill and finding more cleaner alternatives such as recycling or use of biodegradable packages. Sustainability will also be promoted, as the vaccine plant will run on a hydropower dam. However, the build of a hydropower dam would reduce the number of fish and alter the balance of the local ecosystem as fish can no longer swim upstream or spawn in their spawning habitat. To overcome this, fish ladders and spawning simulation can be built to assist fish migration and reproduction. This is not only environmentally sustainable, but also socially sustainable as fishing is an essential industry in Uganda, a reduction of fish in rivers would bring up mass dissatisfaction, hunger and poverty in the Uganda population. Moreover, it is essential to dispose/recycle used materials appropriately, especially plastics. Most vaccination syringes and packaging are made from Polypropylene plastic, although polypropylene can be easily recycled, in Uganda, a large portion of plastic are being dumped into landfills or carelessly thrown on the roadsides, this can potentially lead to serious contamination of national parks or land for agriculture and cattle rearing. To prevent this from happening, waste plastics can be sent back to the manufacturing site for cleaning and sterilisation.

[1] (2016). WHO. *The Ministry of Health and partners review the National TB prevalence survey report*. Available at: https://www.afro.who.int/news/ministry-health-and-partners-review-national-tb-prevalence-survey-report

Once the plastics are sterile, they can be sold to the closest Kabale Plastic Recycling Center (to minimize carbon emission) or local plastic companies for further uses, the sold plastics would then cover up some of its cost of cleaning and sterilisation

Market Research

reIn terms of commercial price, BTBF's aim is to make the vaccine free of cost, as the average individual income of a ugandan citizen is approximately only $170 a month[2]. Finally, the city of Mbarara was chosen for the vaccine plant location as it offers many benefits. The city has several universities which can be used for research, it has close water resource proximity, it has an airport and is connected by roads to the rest of the country. More importantly, it is located near the southern border of Uganda, near countries such as Tanzania, Rwanda, and the Democratic Republic of Congo, where BTBF hopes to expand and help in their fight against TB.

Experimental Set-up

According to the findings of the 3 sub-tasks, the overall design of the bioreactor consisted of a magnetic stirrer (placed in the bottom of the bioreactor), 2 rushton impellers placed 26mm apart on a mixing rod, and three baffles on the wall of the bioreactor (dimension of the baffle can be find in the mixing sub-task). With the aim of having the shortest mixing time while avoiding the formation of vortex, the final agitation speed of the impeller was chosen to be 240 rounds per minute (rpm). For heating, a combination of cotton and aluminium foil was used as an insulator, and the cup was first covered with aluminium foil then cotton. The hotplate was set at 40 °C as we wished for a liquid temperature between 25°C-30°C . Four electrodes were used to measure the voltage of the yeast solution. A drive electrode was "paired" with a sense electrode (placed close to each other), such that the sense electrode immediately receives the signal sent by the drive electrode. The voltages obtained were then read and, using the calibration curve of amplitude vs. frequency (*available in the appendix*), used to find the final estimated concentration of living cells in the bioreactor.

Process Design

After culturing the yeast cells, the cultured product needs to go through a series of unit operations to extract and purify the protein (antibody) from the cultured product. Since the protein is intracellularly produced within the yeast cell, the cultured product needs to first go through an isolation stage consisting of a disc stack centrifuge to separate the denser yeast cell (solid) from the less dense growth medium (liquid). Then, the liquid is disposed of, and the yeast cells are broken open by a homogenizer to release the proteins produced intracellularly. However, at this stage, the homogenised yeast cell also contains cell debris together with the desired proteins, hence, purification stages are also required to filter the desired protein from unwanted products. Microfiltration is used to remove excess water and large unwanted products (cell debris), ultrafiltration is then used to further remove excess water and smaller-sized contaminants. Affinity chromatography is used to remove the remaining contaminants based on its affinity properties. After purification, a pure vaccine (protein/antibody) is obtained for the formulation stage. In this project, the liquid vaccine is spray-dried to increase its protein stability.

[2] Average salary survey. *Uganda/Salary*. Available at: https://www.averagesalarysurvey.com/uganda

Other than the main unit operation (equipment), an air compressor, water for injection (WFI) generator, WFI storage vessel, WFI distribution pump and steam generator are used to facilitate compressed air, WFI and steam into the equipment to conduct the unit operations.

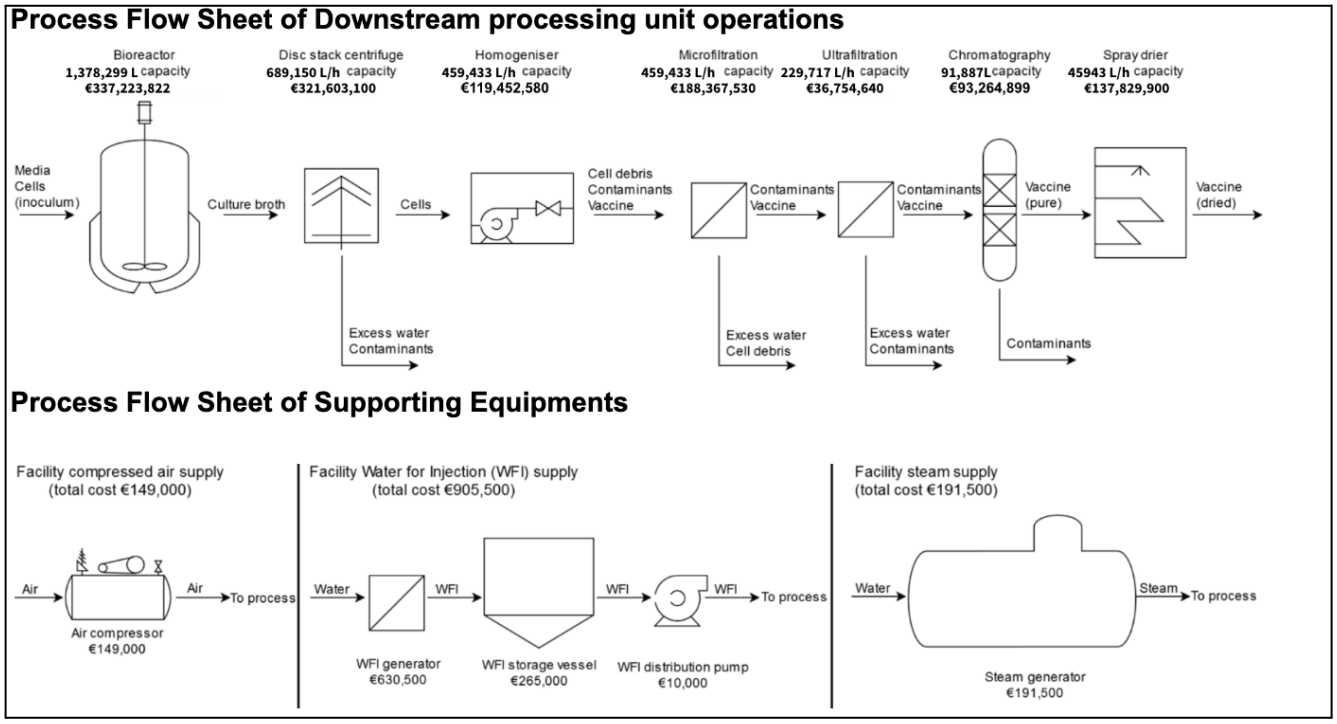

Figure 1. Process Flow Sheet Downstream Processing Unit Operations (top) and of Supporting Elements (bottom)[3]

Estimated Cost of Production

In the initial condition, when the yield of yeast was supposed to be 50g/L the cost of equipments, cost of supporting equipments, physical plant cost and operation cost was calculated to be €841, 925, 771, €1, 246, 000, €4, 444, 187, 295 and €666, 628, 094 respectively. The estimated cost per dose (CpD) for this condition was €46. 05**.** After the 24hr run, the actual yield of yeast was examined to be 34.1g/L. When the yeast yield decreased, the consequent facility cost and cost per dose would increase as more capacity would be required to make the same amount of yeast. Under this condition, the cost of equipments, cost of supporting equipments, physical plant cost and operation cost was calculated to be €1, 234, 496, 471, €1, 246, 000, €4, 444, 187, 295 and €666, 628, 094 respectively. The estimated CpD for the actual 24hr run was then calculated to be €67. 52.

Conclusion

In conclusion, this project will benefit Uganda and its citizens from many aspects. The first and most important one is their health, as the new vaccine will offer a free and better immunisation against TB and hopefully reduce the number of TB cases in the country. Furthermore, the plant could eventually start exporting their vaccine

[3] (2022). ENGF001 Team Challenge. Scale-up and Cost Powerpoint Slide Presentation (Moodle). *The Outline Process Flowsheet.* Slide 8 and 9.

doses to neighbouring countries, therefore contributing to the growth of Uganda's economy. Finally, as the plant will run on renewable energy, the project promotes sustainability.

APPENDICES

1. HEATING

I. Overview of Experiment

The goal of the heating subtask was to hold the reactor contents at a fixed temperature, as temperature plays an important role in determining the production rate and the formation of the desired product. A system focusing on keeping a fixed temperature was designed, using certain heat settings for the plate the cup was placed on and the materials of the external layers wrapping it. Aluminium foil and cotton wool were tested to see which was the best insulator. By the end of the experiment it was found that the layer of aluminium foil and the cotton wool together worked best in preserving the heat inside the plastic cup. The temperature was set at 140 °C and the time needed for the temperature to rise to 28 °C was measured. The aluminium foil and cotton wool layer took 404 seconds to heat compared to the cotton taking 428 seconds. Adding no layers took 739 seconds to reach 28°C. Thus, it was concluded that an aluminium foil and a cotton wool layer combined worked the best.

II. Introduction, Context, and Objectives

Maintaining an appropriate temperature plays an essential role in the production of the yeast. It can be found that the optimal temperature for yeast to grow is between 25°C to 30 °C[4]. Any temperature above or below could either kill the yeast or simply stop its growth. The optimum temperature used for the experiment was considered to be 28°C, as it was the average between 25°C and 30°C. From there, a variety of insulation materials were tested to determine which worked best in keeping a temperature of 28°C in the cup.
The heating theory states that conduction is the process by which heat energy is transmitted through collisions between neighbouring molecules[5]. The rate of heat conduction through a medium can be expressed as stated by Fourier's law:

$$\dot{Q} = -kA\frac{\mathrm{d}T}{\mathrm{d}x}$$

Where Q is the heat flux in W, $\mathrm{d}T/\mathrm{d}x$ is the thermal gradient (˚C) along a direction over a material of thickness (m), k is the conductivity (W m^-1 K^-1) and A is the heat transfer area (m^2).
The law states that heat flux is proportional to the temperature gradient, and the lower the heat conduction Q, the better the insulator. This concept was therefore used to find out which insulator was the best.

III. Materials and Methods

The dependent variable was temperature (measured using a thermometer) and the independent variable was the insulator material (which was changed to obtain different results). The materials used were a digital thermometer, cotton wool, aluminium foil, scissors, tape, and a hot plate.
Multiple experiments were made to find the best insulator without adding the mixing factor. Initially, no insulation was used around the cup, and the temperature on the hotplate was set to 140 °C. This showed a time of 739 seconds for the temperature to reach 28 °C. Then, the experiment was repeated with only cotton wrapped

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[4] Yalcin, S.K. and Yesim Ozbas, Z. (2008) *Effects of ph and temperature on growth and glycerol production kinetics of two indigenous wine strains of saccharomyces cerevisiae from Turkey*, *Brazilian journal of microbiology : [publication of the Brazilian Society for Microbiology]*. U.S. National Library of Medicine.
[5] *ENGF0001 – INTEGRATED ENGINEERING SUBTASK GUIDANCE DOCUMENT*

around the cup. This took 428 seconds this time to reach 28 °C. Finally, the procedure was repeated one last time with aluminium foil and cotton wool. This took only 404 seconds to reach 28 °C. It was therefore concluded that the aluminium foil and cotton wool together were the best insulator.
Subsequently, the mixing factor was added, and the procedure (with aluminium foil and cotton wool) was repeated at a much lower temperature of 35 °C. Over a period of around 1 hour and 40 minutes, the temperature reached 25.2 °C and remained constant after that. Thus, these conditions were used for the 24 hour run.

IV. Results and discussion

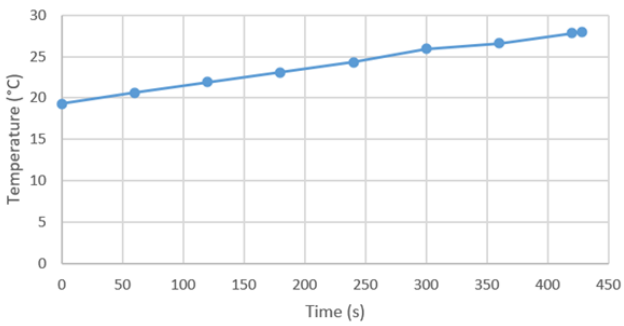


Figure 1. the graph of the temperature of water over time using cotton as insulator (140 °C plate, no mixing)

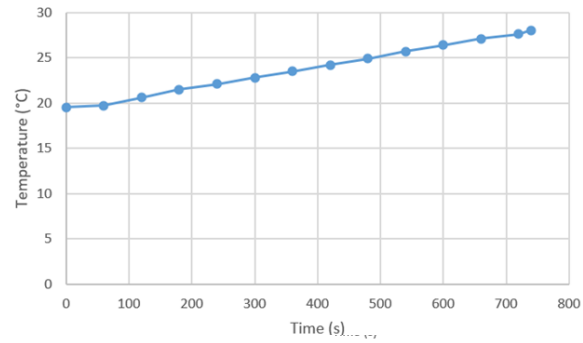


Figure 2. the graph of the temperature of water over time using nothing to insulate (140 °C plate, no mixing)

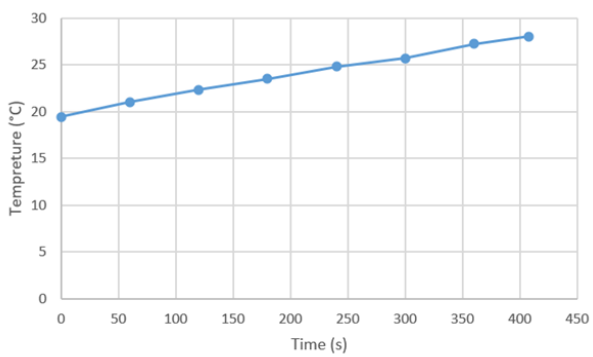


Figure 3. the graph of the temperature of water over time using cotton with foil as insulator (140 °C plate, no mixing)

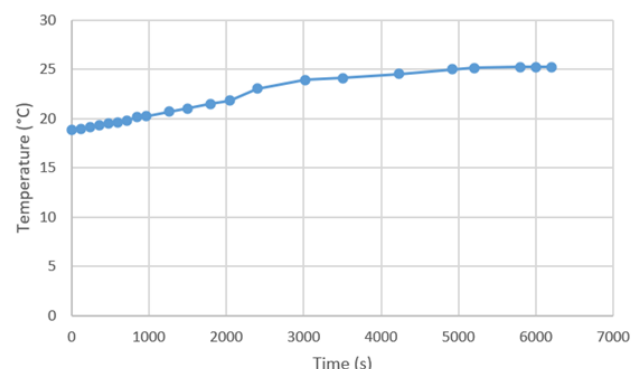


Figure 4. the graph of the temperature of water over time using cotton with foil as insulator (35°C plate, mixing)

Figures 1, 2 and 3, which relate to the first 3 experiments, showed that the combination of cotton and aluminium foil was the best insulator for the 24 hour run, since they took the least amount of time to reach the targeted temperature, all by minimising heat loss. Moreover, figure 4 showed that choosing a hotplate temperature of 40°C during the 24 hour run was optimal. This assumption was made based on the fact that setting the hotplate to 35°C reached 25.2 °C and remained constant. Thus, 40°C should reach around 25°C to 30 °C, closer to 30°C.

Several errors can be taken into consideration for the heating sub-task. One could be related to the fact that the insulator was chosen when a temperature of 140 degrees was tested. The efficiency of the insulator could have changed when the temperature lowered to 35 degrees. Moreover, the temperature was always measured at the surface of the water. This is not always accurate, as there could be temperature differences in different heights of the cup. The temperature at the bottom was probably higher than what was measured. Another potential error could be related to the cotton-only experiment: the bottom of the cup had been covered with cotton, so the hotplate was not directly in contact with the hotplate.

Other problems faced were time and equipment limits. The first experiments were conducted in different water volumes than the ones used in the mixing sub-tasks. This resulted in the need of repeating the experiments again. This can be avoided in the future by enforcing better communication between the heating and mixing. In addition, the heating subtask heavily relied on the mixing group. If more mixing equipment was provided, more time could have been contributed to testing with the impellers instead.

V. Conclusion

In conclusion, our objectives were achieved as a suitable insulator which kept the temperature optimal for the yeast to grow was found. The combination of aluminium foil and cotton wool worked the best. After completing a final experiment at 35 °C, a small temperature increase was seen, and using a higher temperature of 40 °C on the hotplate for the 24 hour run was decided.

2. MIXING

I. Overview of Experiment

The goal of mixing is to reach a homogenous mixing environment in a short amount of time while avoiding the hinder of cell growth and it is characterised by factors including the agitation speed, baffle, type of impeller and number of impellers. By the end of the experiment it was found that the use of two rushton impellers and the highest agitation speed will result in the shortest mixing time. The highest agitation speed was concluded to be 240 rpm because vortex was identified from 180 rpm, and the use of baffles can only inhibit vortex up to 240 rpm.

II. Introduction, Context, and Objectives

When culturing yeasts in bioreactors, it is essential to ensure that all yeast cells are supplied with appropriate and sufficient growing conditions to maximise cell yield. Mixing is also important to the two other sub tasks to distribute temperature evenly throughout the bioreactor and to make sure that the reading of the probe at any location of the bioreactor will represent the entire bioreactor.
According to the mixing time theory, it is ideal to have a short mixing time (t_m) (time taken for the solution to reach homogeneity). Which can be modelled with the equation shown below. As shown, if the agitation speed increases, the mixing time will be shortened, being more ideal. However, it is also important to avoid extreme agitation speeds that lead to high shearing stress or formation of vortex. Although yeasts are low shearing sensitive, high shearing stress can still kill yeast cells. Vortex concentrates yeast cells in the centre of the bioreactor which deviates mixing from reaching homogeneity. Fortunately, baffles can be added to avoid vortex by changing the flow of fluid.

$$t_m = \frac{6.34}{N\left(\frac{D}{T}\right)^{2.3}\left(\frac{Z}{T}\right)^{-0.5}}$$

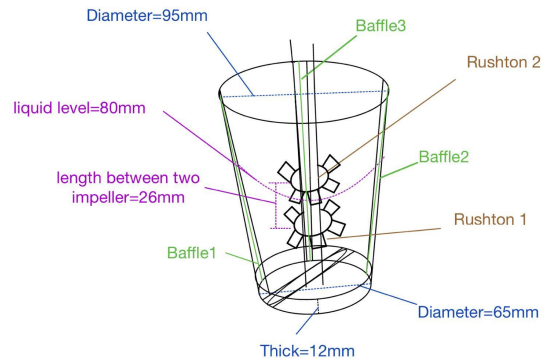
Additionally, considering that air can't be sparged into the bioreactor, all aeration will take place on the surface of the liquid. Hence, mixing was also designed to maximise the surface area to optimise aeration.

III. Materials and Methods

The variables of the experiment are listed below, when one variable was chosen as the independent variable, the remaining variables are controlled, and when the variable is controlled, the underlined condition was used.

- Agitation speed (rpm): 60, 80, <u>100</u>, …, 280 with 20 rpm intervals (with low shearing to not kill yeast)
- Baffles: with baffle, <u>without baffle</u> (use to reduce vortex)

- Impeller type: Rushton (chosen to be fixed as it is the best impeller for high density mixing)
- Number of impeller: 1 impeller, 2 impellers



The diagram to the right shows the experimental setup. 275ml of water was measured with a measuring cylinder. The 275ml water corresponds with 80mm liquid level, which is chosen to be 1:1 to the tank diameter. As the tank diameter is not the same on the top and bottom an average was taken by adding the top and bottom diameters and then dividing it by 2. The distance from the bottom to rushton1 and from Rushton 1 to Rushton 2 are both 26mm, chosen to be the equivalent to the diameter of the impeller.

With the research of ideal baffle dimension, the baffle is made from plastic sheets with a top width of 7.9m (1/12 of the tank top diameter), bottom width of 5.4mm (1/12 of the tank bottom diameter) and length of 85mm (5mm taller than the liquid level). The baffle is also placed 5mm above the magnetic stirrer and 1.1mm (1/6 of the baffle width) away from the tank wall to avoid yeasts being stuck.

A magnetic stirrer is used to rotate the Rushton impeller. When the stirring speed is reached, 3 drops of dye was added and at the same time timing begins. Then, the timer is stopped and time is recorded when the dye is fully dissolved in water.

IV. Experiment result data

As shown in the table of results, the experiment was repeated for 5 trials for greater reliability and an average (highlighted in yellow) of the quantitative data was calculated. Quantitative data was also collected on the presence of vortex.

Number of impellers	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Trial 4 (s)	Trial 5 (s)	Mean (s)	Vortex?
1	19.82	18.3	17.53	20.8	19.34	19.16	no
2	16.58	17.2	16.58	18.98	14.13	16.69	no

Table 2.1: Result (mixing time) when the number of impellers is the independent variable

Table 2: Result (mixing time) when agitation speed is the independent variable (no baffle)

Agitation speed (rpm)	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Trial 4 (s)	Trial 5 (s)	Mean (s)	Vortex?
60	23.08	24.88	24.17	23.81	25.25	24.24	no
80	16.58	17.2	16.58	18.98	15.13	16.89	no
100	12.96	13.34	15.67	11.87	12.6	13.29	no
120	11.48	13.74	12.26	15.53	10.37	12.68	no
140	10.63	12.34	11.47	10.32	10.83	11.12	no
160	10.41	9.39	11.23	10.21	10.58	10.36	no
180	8.41	9.21	8.28	10.1	8.32	8.86	yes
200	8.33	9.18	9.57	9.22	8.71	9.00	yes
220	7.94	8.43	8.2	8.62	7.59	8.16	yes
240	7.54	7.39	8.46	7.67	7.19	7.65	yes
260	6.97	9.56	7.38	7.26	6.98	7.63	yes
280	6.18	7.28	6,89	6.57	7.21	6.81	yes

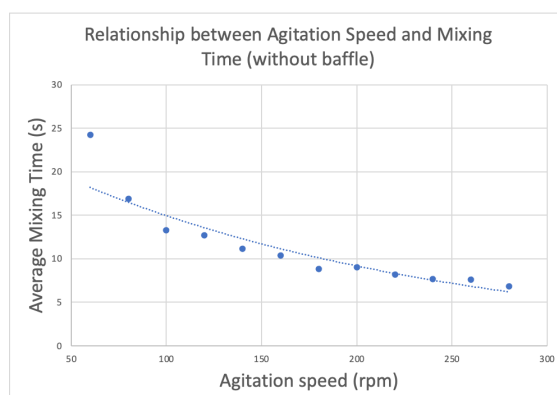
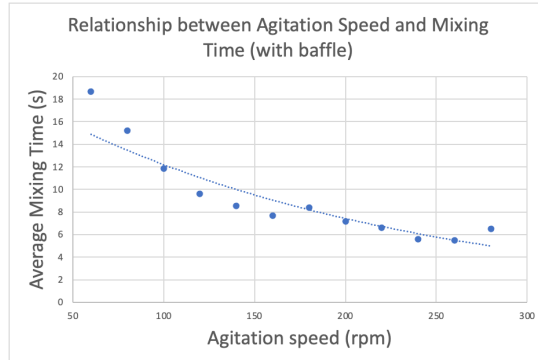


Table 3: Result (mixing time) when agitation speed is the independent variable (with baffle)

Agitation speed (rpm)	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Trial 4 (s)	Trial 5 (s)	Mean	Vortex?
60	17.83	20.11	18.67	18.25	18.45	18.66	no
80	15.15	15.25	15.8	15.01	14.88	15.22	no
100	10.65	11.57	13.42	12.2	11.49	11.87	no
120	9.74	10.87	9.46	9.58	8.48	9.63	no
140	8.03	8.19	9.4	7.39	9.58	8.52	no
160	7.82	7.38	8.49	6.94	7.68	7.66	no
180	8.4	8.49	7.29	9.4	8.28	8.37	no
200	6.31	7.38	7.17	6.93	8.15	7.19	no
220	5.9	6.46	6.38	6.86	7.36	6.59	no
240	5.78	6.29	4.96	5.39	5.47	5.58	no
260	4.98	5.3	5.28	6.12	5.68	5.47	yes
280	4.29	6.21	7.48	6.74	7.73	6.49	yes



According to table 1, the average mixing time with one impeller was 19.16s and with two impellers was 16.69s, with reference to the mixing theory, the use of two impellers would be a more ideal condition as it has a shorter mixing time.

With reference to table 2,3 and the graphs, it is clear to see that there is a strong positive correlation between the increase in agitation speed and the decrease of mixing time. Additionally, when the baffle was not used, a vortex formed when the agitation speed reached 180 rpm. Whereas, when baffles were added to the bioreactor, it effectively avoided the formation of vortexes up to 240 rpm. According to the mixing theory, as a shorter mixing time is desirable and vortex is not favoured, the final agitation speed was chosen to be 240 rpm.

Four experimental limitations were identified throughout the process. Firstly, although the location of dye addition was kept to be consistent, the pace of adding the 3 drops of dye was not carefully controlled. This can lead to errors and reduced data accuracy, to improve, a stop watch can be used to assist the pace of 1 drop of dye per second. Secondly, considering the project as a whole, the experiment could be much more significant if mixing is conducted in fluids with temperatures assigned by the heating group with yeasts in the fluid because temperature will have an impact on mixing. Moreover, mixing liquid dye is also different from mixing solid yeasts. Thirdly, uncertainties might have also resulted from the measuring instruments such as the measuring cylinder (± 2.5ml), stop watch (± 0.05s) and ruler (± 0.5mm). Hence, strictly speaking, data resulted should have a confidence interval of *mean* ± *variation*. Lastly, during the experiment, it was observed that the magnetic stirring block in the bottom of the bioreactor also contributed to mixing dye in the bottom of the bioreactor, consequently questioning the reliability of the data,mixing time could be slightly longer if the magnetic stirring block was removed. To improve, a thinner stirrer block or an overhead stirrer can be used.

V. Conclusion

The objective of the mixing subtask was to find mixing factors that use the least amount of time to reach homogeneity while avoiding the formation of vortex and high shearing. In the end the initial aim was achieved. At 240 rpm, the formation of vortex can be effectively avoided by the researched baffle design. Additionally, it also has a shorter mixing time in comparison to the lower speeds (which does not vortex).

VI. Notation

$$t_m = \frac{6.34}{N\left(\frac{D}{T}\right)^{2.3}\left(\frac{Z}{T}\right)^{-0.5}}$$ [6]

Where, N: $rotational\ speed$, D: $impeller\ diameter$, Z: $liquid\ level$, T: $tank\ diameter$

[6] (2022). ENGF001 Team Challenge. Mixing Subtask Guidance Document (Moodle). Mixing theory equation. Page 1

3. IMPEDANCE PROBE

I. Overview of Experiment

The goal of the task was to build and calibrate impedance probes to monitor the growth of the living yeast during 24 hours. This can be done by measuring the voltage of the yeast concentration inside the bioreactor. In theory, after several test experiments, it was found that inputting an amplitude of 5 Vp-p, a gain of 100, and a frequency of 5,000 Hz was optimal for cell growth monitoring. Moreover, several probe placements were tested, and it was observed that electrodes placed as far as possible from each other recorded higher voltages. Once the settings were obtained, graphs representing the change of amplitude with respect to frequency were plotted for a 20 g/L yeast solution (logarithmic increasing function), then for a PBS solution. Finally, a calibration curve (increasing linear function) representing the change of amplitude with respect to yeast concentration was obtained.

II. Introduction, Context and Objectives

Yeast cells - like any other cells in biology - undergo the process of mitosis, where they continuously divide and multiply when placed in optimal growth environments. In the context of the new TB vaccine, yeast cells produce the needed antibodies for the vaccine. By measuring the growth of these cells dynamically, they can be monitored such that they don't reach a death phase. The dynamical measurements can be done using electrical impedance theory, which states that when cells grow, more space in the bioreactor is being used, and extracellular area decreases. Since the extracellular area is inversely proportional to resistance: the smaller the extracellular area, the higher the resistance. According to Ohm's Law, the higher the voltage, the higher the resistance. Thus, the fundamental objective of this experiment is to use this theory, which implies that obtaining higher voltage readings indicate higher cell concentrations (and therefore more vaccine antibodies).

III. Materials and method

To begin, several probe settings needed to be determined using two sense electrodes, two drive electrodes, a PCB circuit, and myDAQ's oscilloscope and function generator. The settings were tested by fixing all parameters, and changing only one variable at a time. The first tested variable was the probe placement. This was done by fixing the amplitude to 5 Vp-p, the gain to 100, and the frequency to 600 Hz. Three placements were tested, which can be found in Figure 3.1 below.

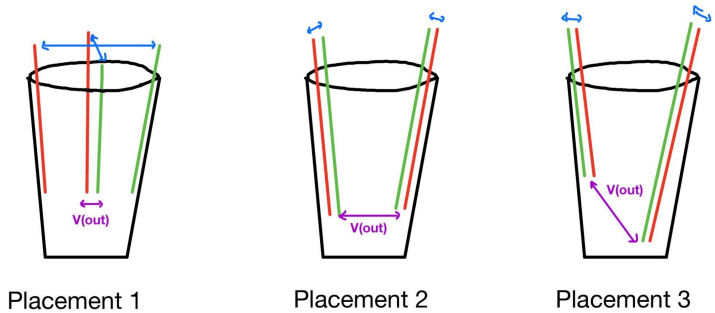


Figure 3.1. Probe placement options

Using the same concept mentioned previously, the second tested variable was the gain and the third was the amplitude.

IV. Results and Discussion

Below is a table summarising the tests and their corresponding results:

Probe placement	Amplitude	Gain	Frequency	Voltage out (in mV)
1	5	100	600	160
2	5	100	600	320
3	5	100	600	603
3	5	0.1	600	4
3	5	1	600	15
3	5	10	600	75
3	5	100	600	376
3	5	1000	600	4,480
3	1	100	600	76
3	5	100	600	372
3	10	100	600	738
3	10	100	1	N/a
3	10	100	10	N/a
3	10	100	100	Results were too small
3	10	100	500	Started obtaining readings

Figure 3.2. Table representing the tests that were conducted. Probe tests are in grey, gain tests are in green, amplitude tests are in blue, and the starting frequency range is in orange

Probe placement was seen to be optimal when placed in position 3, as the first drive electrode (sending the signal) was very close to the first sense electrode (receiving the signal). Since current spreads out like a magnetic field, placing the first pair (drive/sense) as far as possible from the second pair showed higher voltage readings. In terms of gain, any value under 100 was not providing voltages which were high enough, and the higher the gain, the higher the voltage in the solution. The similar effect was seen with amplitude: as it was increasing, signals became harder to read. However, there is a limit of gain and amplitude the PCB can handle, which - if exceeded - can cause saturation. This means the system becomes overloaded, and signals become distorted and inaccurate[7]. Once the parameters were set, frequencies starting at 500 Hz were used, and plots for yeast and PBS, as well as a calibration curve were obtained. (Figure 3.3)

The curve representing the behaviour of PBS was supposed to be flat and linear. This could be linked to the fact that another electrical component was measured instead (amplifier or filter), and should have been improved. Nevertheless, the final frequency was estimated based on the largest voltage difference between yeast and PBS, where the PBS line was stable. Thus, a frequency of 5,000 Hz was chosen, where the amplitude was 497mV.

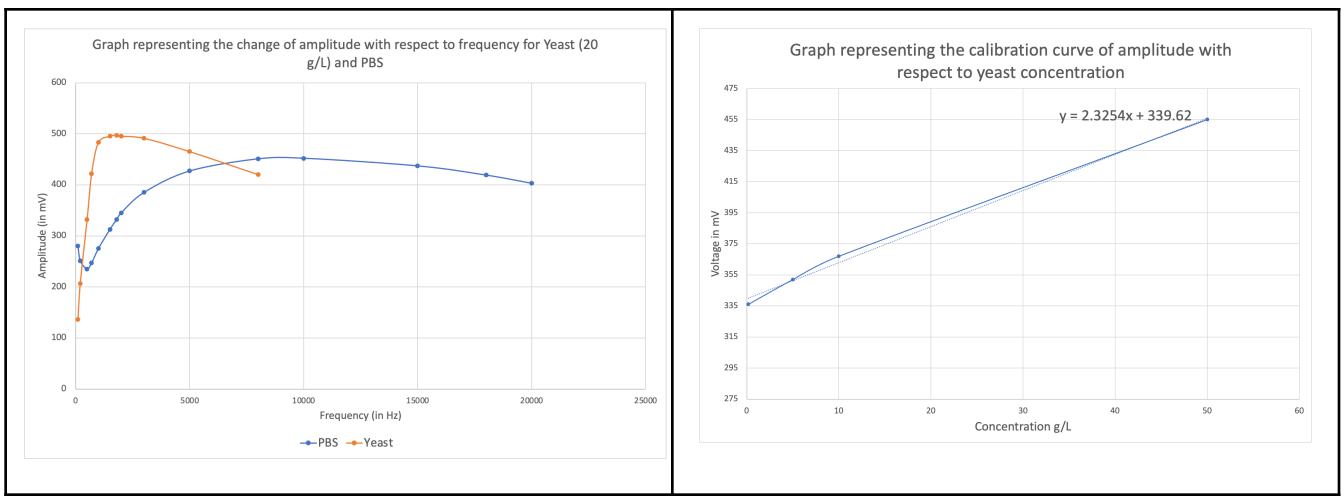


Figure 3.3. Amplitude vs. Frequency plots for Yeast and PBS (right) and Calibration Curve

V. Conclusion

[7] Mastering. Sage Audio. *What's the difference between distortion and saturation?* Available at: https://www.sageaudio.com/blog/mixing/whats-the-difference-between-distortion-and-saturation.php#:~:text=Saturation%2C%20when%20talking%20about%20audio,components%20of%20an%20electrical%20system.

In conclusion, the yeast plot met the initial goal, as it revealed a maximal voltage of around 500 mV. The PBS plot however, was not very accurate, and leaves room for improvements. Nevertheless, a frequency of 5,000 Hz was estimated to be coherent with the remaining parameters - gain of 100, amplitude of 5 Vp-p, and probe placement 3 - which were the final choices made for the design of the bioreactor.

4. THE 24 HOUR RUN

The 24 hour run was the final test which determined the effectiveness of the bioreactor design. The run started at 12pm, and used the findings established by the mixing, heating, and probe teams. Initially, 2 g/L of yeast cells was put into the reactor to test cell growth. Throughout the experiment, three sources of data were collected. The impedance probe was used to determine the concentration of living cells in the bioreactor, the absorbance value of the UV spectroscopy was used to determine the concentration of total cells (living and dead cells) and a dry cell weight to suggest the weight biomass contained in the bioreactor.

The probe calibration curve's equation (acquired from the probe subtask) $y = 2.3254x + 339.62$ was used to convert the probe's voltage reading into the concentration of yeast cells.

Experiment Duration (h)	0	1	2	3	4	5	20	21	22	23	24
Probe Reading (mV)	445.196	445.385	427.37	444.088	483.95	534.694	479.828	485.698	487.252	488.496	487.711
Calibrated Concentration (g/L)	45.401	45.482	37.735	44.925	62.067	83.888	60.294	62.818	63.487	64.022	63.684

Table 4.1: Data collected from the impedance probe

According to the probe data, at the end of the 24 hour run, 63.684 g/L of living cells were expected in the bioreactor.

The total number of cells was determined using the absolute absorbance and turbidity measured by UV spectroscopy. During the data collection process, the concentration of yeast was diluted with water using different dilution factors to keep the absolute absorbance value between 0.2 to 0.8 for optimum accuracy. *The calibrated concentration was obtained using a calibration curve with the equation of $y = 1.1173x$.

Time	0	1	2	3	4	5	6	20	21	22	23	24
Water	700	700	700	800	800	800	800	800	800	900	900	900
Yeast	300	300	300	200	200	200	200	200	200	100	100	100
Absorbance	0.646	0.785	0.822	0.603	0.489	0.296	0.288	0.391	0.777	0.55	0.511	0.548
Calibrated concentration*	0.57818	0.70259	0.7357	0.53969	0.43766	0.26492	0.25776	0.34995	0.69542	0.49226	0.45735	0.49047
Final concentration	1.927266667	2.341966667	2.452333333	2.69845	2.1883	1.3246	1.2888	1.74975	3.4771	4.9226	4.5735	4.9047
Turbidity	22.6	16.4	26.8	13.1	32.4	50.6	51.5	40.6	16.7	28.2	30.8	28.3

Table 4.2: Data collected from the UV spectroscopy

The data in table 5 suggests that the total cell concentration of yeast was 4.9047 g/L. However, the final dry yeast cell weight was examined as 34.1 g/L.

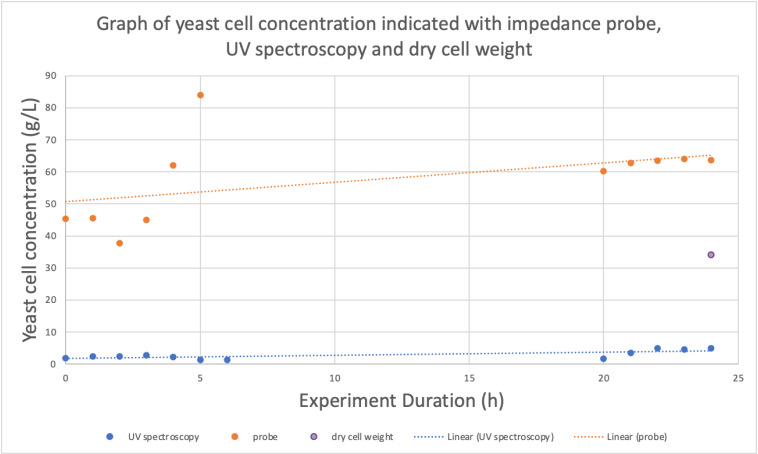


Figure 4.3: Graph of yeast cell concentration indicated with impedance probe, Uv spectroscopy and dry cell weight

As shown in the graph, the positive sloping trendline of the probe and UV spectroscopy reading suggests that yeast in the bioreactor is gradually growing. It is also noticeable that the data collected by the probe follows the typical cell growth stages. In the first 2 hours, the yeast cell is in the lag phase, where there is not a lot of cell growth and even with a decline in cell number. From the second to the fifth hour, the yeast cells were in the lag phase, multiplying in numbers rapidly. Then in the last 5 hours the cell reaches the stationary stage, where cell growth begins to crease.
The UV spectroscopy data, however, does not follow the typical cell growth stage as supposed. Cell numbers began to decrease after the third hour. During the experiment, it was observed that mixing did not reach homogeneity, solutions in the bottom of the bioreactor had a higher turbidity in comparison to the top, this suggests that most of the yeast cells had sunk to the bottom due to their density. Additionally, since one team member extracted the yeast solution from the bottom of the bioreactor, while others extracted it from the top, the solution was more diluted in some measurements than in others, therefore providing inaccurate absorbance readings.

Moreover, as shown in the graph, the yeast cell concentration obtained in all three methods varies greatly. Errors could have occurred in the probe measurements due to electrode movements or time limitations (measurements taken in a rush). This could have resulted in an inaccurate probe calibration curve. However, assuming that the probe reading has the greatest accuracy (as it follows the typical cell growth stage) the deviation of the dry cell weight and UV spectroscopy can be explained with two assumptions. The lower yeast concentration determined by dry cell weight (34.1g/L) in comparison to probe (64.7g/L) can be explained by the removal of water mass when the cell is dried, considering that a large portion of the yeast cells are water, the removal of water can significantly reduce the weight of the cell. The deviation of the UV spectroscopy data could be a consequence of the errors mentioned in the previous paragraph, or alternatively a result of an inappropriate use of the calibration curve (as the calibration curve used was not specialised for yeast cells).

5. CALCULATIONS AND ASSUMPTIONS

I. Calculation of the Required Bioreactor Capacity for the initial and final 24hr run.

The initial condition when the yield of yeast was 50g/L

In general, the mass of protein (g) needed can be calculated as:

$$94,000,000\ (number\ of\ doses\ needed)\ x\ 50\mu g\ (grams\ of\ protein\ needed\ per\ dose)\ =\ 4700g$$

And because 50g of yeast is produced per litre and only 0.01% (w/w) of the yeast is the actual targeted protein (antibody), so only 0.005g of protein is produced per L. To obtain 4700g of protein, the capacity of the bioreactor can then be calculated by:

$$4700(mass\ of\ protein\ required)\ \div\ 0.005(amount\ of\ protein\ obtained\ per\ litre)\ =\ 940,000L$$

<u>The 24hr run condition when the yield of yeast was 34.1g/L</u>

The mass of protein (g) needed can be calculated as:

$$94,000,000\ (number\ of\ doses\ needed)\ x\ 50\mu g\ (grams\ of\ protein\ needed\ per\ dose)\ =\ 4700g$$

And because now only 34.1g/L of yeast is produced in the actual 24hr run, and 0.01% (w/w) of the yeast is the actual targeted protein (antibody), so only 0.00341g of protein is produced per L. To obtain 4700g of protein, the new capacity of the bioreactor can then be calculated by:

$$4700(mass\ of\ protein\ required)\ \div\ 0.00341(amount\ of\ protein\ obtained\ per\ litre)\ =\ 1,378,299L$$

II. Calculation of the Cost of Large-scale process for the Initial and Final 24hr Run

<u>The initial condition when the yield of yeast was 50g/L</u>

To calculate the cost of the large-scale process, the cost given by a 3000L bioreactor will first need to be scaled-up to a bioreactor with a capacity of $940,000L$.

The scale up factor can be calculated as:	$\frac{940,000}{3000} = 313.33$	

The cost of equipment of a 3000L bioreactor is €$2,687,000$, considering that the cost of the equipment required can be scaled linearly with size of bioreactor, the cost of equipment of a $940,000L$ bioreactor can be calculated

	$2,687,000 \times 313.33 = 841,925,771$	

The cost of supporting supply for a 3000L bioreactor is €$1,246,000$, since the cost of supporting supply does not change when the capacity of the bioreactor changes, the cost of supporting supply of a $940,000L$ bioreactor is still €$1,246,000$.

The physical plant cost (PPC) can then be calculated by multiplying the cost of equipment by a Lang factor (shown in the figure below). There are 3 Lang factors, in the end, the Lang factor for a mixed fluids-solid process was chosen because the bioreactor of the project includes liquid growth medium and solid yeast cells.

$f_L = 3.1$ for predominantly solids processing plant
$f_L = 4.7$ for predominantly fluids processing plant
$f_L = 3.6$ for a mixed fluids-solids processing plant [8]

The PPC is then calculated:

[8] Coulson and Richardson, Coulson and Richardson's Chemical Engineering Volume 6 - Chemical Engineering Design (4th Edition), page 251.

	$841,925,771 \times 3.6 = 3,030,932,772$	

The annual operating cost is assumed to be 15% of the PPC, and is calculated to be:

	$3,030,932,772 \times 0.15 = 454,639,915$	

Finally, the cost per dose (CpD) is calculated with the formula:

	$CpD = \frac{fixed\ cost + operating\ cost}{number\ of\ dose}$	
	$\frac{843,171,771+3,030,932,772+454,639,915}{94000000} = €46.05$	

Hence, the CpD is calculated to be €46.05.

<u>The 24hr run condition when the yield of yeast was 43.1g/L</u>

With the same process the cost of equipments, cost of supporting equipments, physical plant cost, operating cost and operation cost of the 24hr run condition is calculated to be:

Cost of equipments: $1,234,496,471$

Cost of supporting supply: $1,246,000$

Physical plant cost: $1,234,496,471 \times 3.6 = 4,444,187,295$

Operating cost: $4,444,187,295 \times 0.15 = 666,628,094$

Cost per dose: $\frac{1,235,742,471+4,444,187,295+666,628,094}{94000000} = €67.52$

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